CLIP-MoA: Visual-Language Models With Mixture of Adapters for Multitask Remote Sensing Image Classification

Zhongzheng Fu, Hongping Yan, and Kun Ding

***Abstract*— In recent years, the rapid growth of remote sensing (RS) datasets has greatly enhanced the difficulty of RS image classification tasks. Given the remarkable success of large vision-language models (VLMs) in various domains, their potential for RS image classification tasks has attracted significant attention. In this study, we focus on contrastive language-image pre-training (CLIP), a widely recognized contrastive learning model known for its strong zero-shot transferability. However, the domain gap between the pre-training data (e.g., natural images) of CLIP and RS data can significantly hinder the application of CLIP. First, the scarcity of labeled RS data poses a significant challenge for fine-tuning CLIP. Second, the complexity and diversity of RS images substantially increase the fine-tuning time and computational resources needed to individually adapt CLIP to new tasks. Third, independently fine-tuning CLIP for each RS dataset ignores the inter-task relationships that frequently exist among RS images from various sources, leading to reduced classification accuracy. To address these challenges, we propose a learnable mixture of adapters (MoAs) module in the visual branch of CLIP, resulting in the CLIP-MoA framework. Inspired by the mixture of experts (MoEs), the MoA module models inter-dataset relationships by integrating multiple adapters through a gating network, while preserving original information via a residual connection between the MoA input and the fine-tuned features. This module enables CLIP-MoA to effectively capture both inter-task commonalities and task-specific characteristics by fine-tuning a minimal set of MoA parameters across multiple datasets simultaneously, thereby reducing computational overhead and fine-tuning time while improving classification accuracy. Comprehensive experiments conducted on a multitask benchmark involving 11 RS image classification datasets demonstrate that CLIP-MoA achieves significant improvements in both classification accuracy and generalization capability compared to zero-shot CLIP, linear probe CLIP, CLIP-Adapter, and CoOp.**

***Index Terms*— Contrastive language-image pre-training-mixture of adapters (CLIP-MoAs), multitask learning (MTL), remote sensing (RS) image classification, vision-language models.**

I. Introduction

RECENTLY, remote sensing (RS) technology has undergone significant advancements, resulting in the development of diverse methods for acquiring RS images with substantial improvements in spectral, spatial, and temporal resolutions. These technologies include aerospace RS, aerial RS, and ground RS, encompassing traditional satellite and aerial imaging, as well as emerging methods such as high-resolution satellites and digital Earth platforms. Different acquisition methods yield distinct RS image datasets with different image features. These datasets have significantly broadened the applications of RS image recognition technology [1]. A prominent example of this is Google Earth [2]. Google Earth utilizes high-resolution RS images and advanced image recognition techniques to offer detailed representations of the Earth's surface and a global 3-D model. These capabilities support educational initiatives, rapid analysis, and facilitate access to extensive RS datasets, enabling scientific research and environmental monitoring.

The evolution of RS image recognition has transitioned from traditional machine learning methods, such as support vector machines (SVMs) [3] and feature-based classifiers, which were once the cornerstone of the field, to deep learning-based neural network methods [4]. Traditional methods relied on manually engineered feature extraction and, although effective in specific scenarios, often demanded substantial domain expertise and manual intervention. With the advent of deep learning, neural network-based approaches [5] have dominated the field of RS image recognition, providing end-to-end solutions and minimizing the need for manual feature engineering. However, these methods require a substantial number of training samples and often lack generalization capabilities, limiting their effectiveness in diverse RS image classification tasks. To address these limitations, domain adaptation (DA) methods have been proposed. For instance, MAOSDAN [56] developed a multiadversarial learning framework for open-set DA to mitigate domain shift in RS scene classification; HyUniDA [57] overcame label set constraints in cross-scene hyperspectral image classification through a universal DA approach; BAN [58] proposed a noise-annotation-based universal paradigm for robust cross-scene classification of RGB and hyperspectral RS images with noisy annotations. While DA provides a promising solution for RS image classification, its practical implementation faces challenges including data heterogeneity, temporal dynamics, and existing methodological limitations.

Recent advances in vision-language models (VLMs), such as contrastive language-image pre-training (CLIP) [22] and

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VisualGPT [18], have revolutionized RS image classification [12], [13], [14], [15], [16], [17] by synergistically combining visual perception with textual semantics. Unlike conventional deep learning approaches that rely solely on low-level pixel features, VLMs incorporate high-level semantic context through natural language processing, enabling more flexible and interpretable classification.

VLMs [8] typically comprise visual feature extractors, such as convolutional neural networks (CNNs) [9] or Transformers [10], Transformer-based language feature extractors (e.g., BERT [6] or GPT [7]), and visual-linguistic feature fusion modules [11]. Furthermore, VLMs may integrate task-specific additional layers to accommodate diverse applications. This flexible architecture allows VLMs to adapt to a wide range of application scenarios, providing capabilities that span from basic visual-textual comprehension to complex generative tasks. Early research relied on manually designed image descriptors, pre-trained word vectors, or frequency-based TF-IDF features. In contrast, recent studies predominantly employ Transformer-based image and text encoders to learn features separately or jointly, combined with strategic pre-training objectives, enabling these models to perform various downstream tasks on large-scale datasets.

Based on their training methodologies and objectives, VLMs can be classified into two categories: generative VLMs, such as VisualGPT [18], RS-GPT4V [19], EarthGPT [16], and bootstrapping language-image pretraining (BLIP) [20], and contrastive learning VLMs [21], including CLIP [22], RemoteCLIP [23] and ProText [24]. Generative VLMs are designed to generate text descriptions corresponding to visual content, whereas contrastive learning VLMs employ self-supervised learning with contrastive strategies on a large-scale dataset of image–text pairs during pre-training.

As a prominent contrastive learning VLM, CLIP leverages 400 million text–image pairs for self-supervised contrastive training, enabling its text and image encoders to encapsulate extensive empirical knowledge. Consequently, CLIP exhibits robust zero-shot capabilities across diverse domains and achieves significant success in visual tasks through the use of hard prompt text inputs. However, additional training or fine-tuning on smaller datasets remains necessary for task-specific applications. To date, a wide range of CLIP-based VLMs and fine-tuning methods have been proposed for various natural image processing tasks. For instance, ProText [24] improves the generalization capability of CLIP by learning prompts exclusively from text data, which can then be transferred to new datasets and categories.

Despite CLIP's remarkable performance across various domains, its application in RS image processing encounters significant challenges owing to the diversity and complexity of RS datasets. The domain gap between the pre-training data (e.g., natural images) and RS data can hinder the effectiveness of fine-tuning. The scarcity of labeled data in RS poses a significant challenge for fine-tuning. RemoteCLIP [23] introduced a data scaling method to mitigate the scarcity of pre-training data in RS domain, and constructed the VLM fundational model of RemoteCLIP by pre-training CLIP on the enlarged pre-training dataset. However, the domain deviations also exist between different RS datasets, including variations in geographical location, sensor imaging methods, and weather conditions, leading to a significant drop in accuracy when a model trained on one dataset is applied to another.

Moreover, these challenges not only impact the model's generalization ability but also result in decreased performance across diverse datasets. To tackle these issues, researchers have developed various approaches to enhance the model's adaptability and generalization. These approaches employ techniques such as DA and multitask learning (MTL) to address the complexities in RS image classification. However, these methods often require extensive computational resources and time, particularly when dealing with a wide variety of RS datasets. As a consequence, there is a need for a fine-tuning method that allows large VLMs to adapt to multiple RS datasets or tasks simultaneously. The primary approach to adapting large models for downstream tasks involves training and storing specific model parameters for each individual task, a strategy known as single-task learning (STL). Alternatively, MTL [25] can be employed.

In deep learning, STL and MTL constitute two dominant paradigms, characterized by distinct implementation strategies and application objectives. STL focuses on addressing a single, well-defined problem, aiming to train a model for optimal performance on that specific task. Existing STL approaches typically necessitate storing a separate set of model parameters for each task, resulting in substantial resource consumption. Furthermore, STL fails to account for inter-task relationships, and individually fine-tuning models for each task demands considerable computational resources. In contrast, MTL aims to handle multiple related or unrelated tasks using a unified model. Its primary advantage lies in leveraging potential inter-task correlations within a single model, replacing the need for multiple models, thereby reducing computational costs while improving learning efficiency and generalization capabilities.

Nevertheless, existing MTL methods continue to encounter several practical challenges in the RS image classification domain. On one hand, many methods remain rooted in STL principles and fail to adequately model inter-task relationships, thereby limiting their performance in multitask scenarios. On the other hand, while MTL techniques can effectively capture task relationships, it remains a significant challenge to achieve a balance between universality and specificity across different RS image classification datasets due to their various resolutions and scalability. Although different RS image classification datasets may share similar category labels, the image features of land cover under the same label can vary significantly across datasets with different resolutions and features. For example, high-resolution satellite images offer detailed texture information, whereas aerial photography can capture more precise spatial relationships, as depicted in Fig. 1. Furthermore, RS images are characterized by large data volumes and high complexity, making their processing computationally intensive in terms of time and memory.

The integration of MTL with pre-trained VLMs has emerged as a promising approach for addressing challenges in the

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(a) (b)

Fig. 1. Examples of RS images derived from different sources. (a) High--resolution satellite image, which can provide detailed texture information. (b) Aerial photograph that can capture more precise spatial relationships.

RS domain. Pre-trained VLMs like CLIP, with their robust zero-shot capabilities and flexible architectures, are ideal for MTL. Combining MTL techniques with VLMs enhances computational efficiency, models inter-task relationships more precisely, and enables effective knowledge transfer. In recent years, considerable progress has been made in applying VLMs and MTL to RS. GeoChat [14], the first versatile VLM for RS, is capable of processing high-resolution RS images and offers multitask conversational capabilities. Di Wang et al. [55] investigate the multitask pretraining (MTP) paradigm for RS foundation models to mitigate task discrepancy when transferring natural image pre-trained models to downstream RS tasks. Using a shared encoder and task-specific decoder architecture, the researchers perform multitask supervised pre-training on the segmentation RS dataset, SAMRS, which includes semantic segmentation, instance segmentation, and rotated object detection. These advancements provide valuable insights and directions for future research in RS.

In this study, we introduce a novel MTL-based parameter-efficient fine-tuning method for CLIP, termed CLIP-MoA, designed to efficiently process multiple downstream RS image classification datasets concurrently. The primary contributions of this work are summarized as follows.

1) We propose a module called mixture of adapters (MoAs), which incorporates the MTL mechanism of mixture of experts (MoEs) and adapter tuning. MoA integrates multiple adapters via a gate mechanism to associate and enhance ground features sharing the same category labels across different RS image classification datasets. Additionally, to preserve the original information, a residual connection is employed between the input of MoA and the fine-tuned image features.
2) We integrate the MoA module into the visual branch of CLIP, termed CLIP-MoA. CLIP-MoA is capable of processing multiple RS image classification datasets concurrently, achieving higher precision, lower computational costs, and enhanced generalization ability by fine-tuning only a small number of parameters of the MoA module while keeping the backbone of CLIP frozen.
3) We conduct extensive experiments on a multi-task dataset comprising 11 collected RS image classification datasets. Experimental results demonstrate that CLIP-MoA achieves superior performance with higher efficiency on multiple RS datasets simultaneously with less computational costs compared to zero-shot CLIP, linear probe CLIP, CLIP-Adapter, and CoOP.

II. Related Work

A. RS Datasets

Currently, RS data encompasses a diverse range of sources, including high [26], medium, and low [27] resolution imagery, multispectral [28], hyperspectral [29], synthetic aperture radar (SAR) [30], street view, and LiDAR point cloud data. These data sources provide foundational support for RS monitoring and interdisciplinary applications.

Aerial photography datasets [31], renowned for their high resolution and precision, are capable of capturing fine surface details, making them suitable for small-scale, high-resolution analysis. They are extensively utilized in urban planning, agricultural monitoring, and disaster assessment. For example, NASA's Aerial Imagery for Research Program (AIRP) provides a vast collection of aerial imagery data, supporting research on land cover and land use change analysis.

Satellite RS datasets [32] are highly valued for their extensive coverage and regular updates, making them ideal for long-term global and regional monitoring. The Landsat series of satellites [33] and Sentinel-2 satellites provide high-resolution multispectral images, which are widely recognized as classic RS sources and are extensively applied in agriculture, forestry, and environmental monitoring [34], [35].

B. Multitask Learning

The approaches to MTL are broadly categorized into two main types: hard sharing and soft sharing. As the most common approach, hard sharing involves sharing all model parameters across tasks at lower layers, followed by task-specific differentiation at higher layers. This strategy requires the model to learn a shared representation that accommodates all tasks simultaneously, thereby achieving efficient resource utilization, facilitating knowledge transfer, and mitigating overfitting across tasks. Hard sharing is particularly prevalent in image processing tasks, where the abstract nature of low-level features, which are generally less task-specific, often enables effective results through this low-conflict sharing strategy. Additionally, this approach significantly reduces the computational burden associated with multiple tasks. However, a notable limitation of the hard sharing approach lies in its inherent constraints. The mandatory shared layers often struggle to learn representations that are equally effective for all tasks, leading to suboptimal performance owing to the inability to fully exploit inter-task correlations.

Unlike hard sharing, soft sharing involves partial sharing of lower-level parameters while retaining task-specific parameters. The primary advantage of this approach is its ability to offer task-specific customization. Representative examples of soft sharing include MoEs [42] and multigate MoEs (MMoE) [43], both introduced by Google. These methods partition

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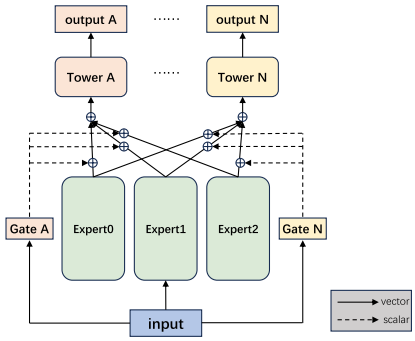


Fig. 2. Structure of MMoE [43].

shared layers into multiple experts and employ gates to dynamically regulate gradient contributions from different experts to downstream tasks. The key distinction between the two lies in the number of gate networks employed. MoE utilizes a single gate network to weight and aggregate the outputs of multiple expert networks, enabling task-specific outputs for different downstream tasks. MMoE, as illustrated in Fig. 2, employs multiple gate networks, each corresponding to a specific downstream task, to interact with multiple expert networks, thereby enhancing feature extraction for individual tasks. Thus, MoE can be viewed as a specialized instance of MMoE.

C. Fine-Tuning Methods

Despite the remarkable zero-shot transfer learning capabilities of VLMs, pre-trained VLMs still necessitate task-specific fine-tuning with labeled training data for each downstream task. During fine-tuning, the model's weights are updated to optimize performance for the target task. As model sizes increase, the computational cost of fine-tuning escalates, driving the development of various lightweight fine-tuning methods, including linear probing, prompt tuning [36], and adapter tuning [37].

Linear probing updates solely the final linear layer (typically the classification layer) of the model, while keeping the weights of other layers fixed. However, models fine-tuned via linear probing often demonstrate limited generalization performance. Prompt tuning and adapter tuning do not necessitate modifying the parameters of pre-trained models. Prompt tuning adapts to new tasks by altering input visual prompts or hard/soft textual prompts. For instance, CoOp [38] utilizes continuous learnable prompts, comprising a randomly initialized soft token integrated with class names.

Adapter tuning incorporates a small number of trainable adapter layers into the model. Typically, the adapter module is a nonlinear projection block with significantly fewer learnable parameters compared to the pre-trained model. During adapter tuning, only a small number of the adapter parameters are updated while keeping the pre-trained model parameters frozen, enabling models to adapt to different tasks while minimizing fine-tuning costs and enhancing efficiency. The introduction of CLIP-Adapter provides a new perspective on fine-tuning VLMs. CLIP-Adapter [39] integrates adapter modules into both the language and image branches, enabling CLIP to adapt to downstream tasks. It achieves strong performance in STL on natural image datasets. However, to address multiple tasks, it is typically required to train a separate model for each task. While this approach is generally feasible, in specific scenarios, MTL demonstrates notable advantages. For example, however, its need for independent adapter training for each RS dataset leads to increased computational costs and fine-tuning time, owing to the unique and specialized nature of RS images.

The NLP community has explored MTL-based parameter-efficient transfer learning methods. AdapterFusion [40] is an enhanced adapter tuning approach that aims to maximize task transfer effectiveness by integrating multiple adapter modules. AdapterFusion divides the learning process into two stages: the knowledge extraction stage and the knowledge combination stage. During the knowledge extraction stage, a new set of adapter parameters is learned for each task. In the knowledge combination stage, a fusion module is learned to combine all adapters from the first step for a specific target task. The fusion module employs an attention mechanism similar to cross-attention in transformers to weight and integrate the outputs of adapters from different tasks. This approach addresses the issues of catastrophic forgetting, inter-task interference, and unstable training. However, the addition of the adapter module increases the overall parameter count, degrading inference performance and significantly prolonging the fine-tuning process.

AdaMix [41] aims to enable efficient model parameter tuning by combining multiple adapters. Inspired by sparsely activated mixture-of-experts, AdaMix employs multiple adaptation modules to learn diverse representations of the given task, treating each adapter as an expert. It utilizes random routing or weight averaging to select adapters during training and leverages these adapters for comprehensive decision-making during inference. Specifically, AdaMix randomly selects an expert from the feedforward up-project layers to process the input and then passes the output to an expert randomly chosen from the feedforward down-project layers after applying a nonlinear activation. This hybrid adapter mechanism enhances the model's adaptability and generalization ability across different tasks. However, this structure intensifies the coupling between expert knowledge.

Different from AdapterFusion and AdaMix, CLIP-MoA introduces the MoA module on the visual branch of CLIP for modeling the relationships among multiple RS datasets. Inspired by the multitask learning mechanism of MoEs, the MoA module adopts a similar architecture as MoE, which parallelizes multiple adapters through a gating mechanism and preserves the input features via a residual connection between the adapter inputs and the gate outputs. Each adapter is regarded as an expert module, learning task-specific features. The residual connection and gating mechanism prevent catastrophic forgetting and facilitate the modeling of inter-task relationships. Compared to AdapterFusion and AdaMix, CLIP-MoA is simpler yet more efficient, enhancing not only classification accuracy but also the model's generalization capability and learning efficiency. It facilitates inter-task information sharing and simultaneously handles multiple RS image

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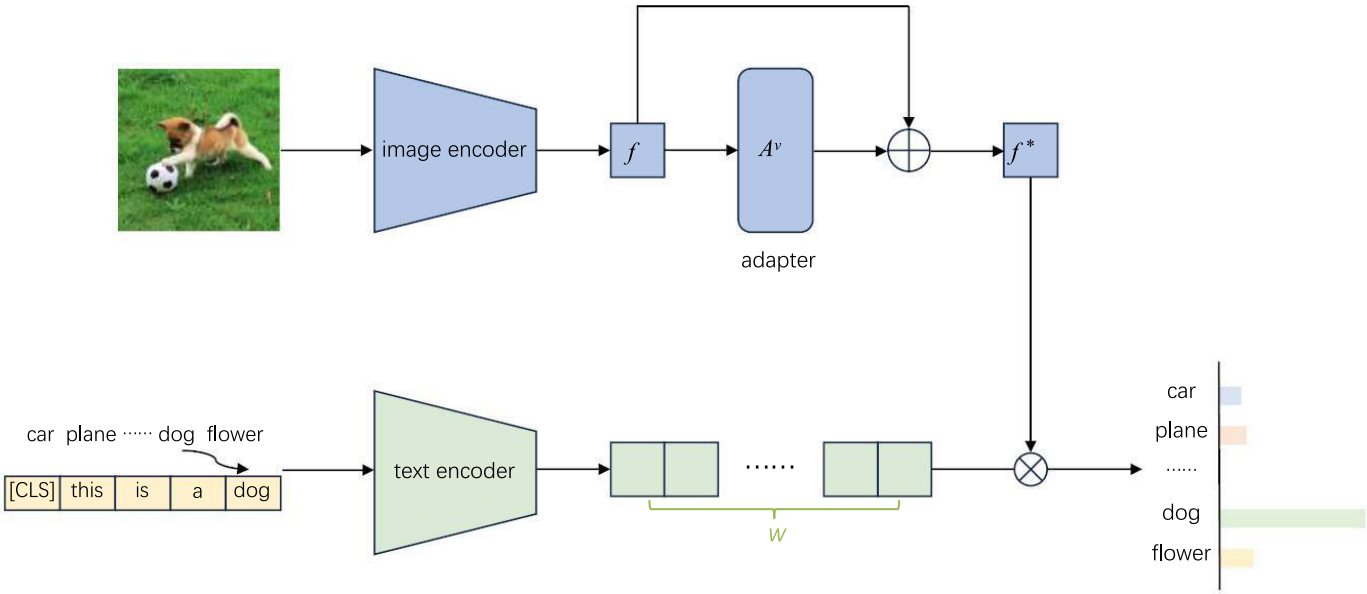


Fig. 3. CLIP-Adapter with one adapter on the image branch.

classification datasets, significantly reducing computational costs.

III. Method

We propose CLIP-MoA, inspired by the MTL mechanism of MoE and the adapter tuning method of CLIP-Adapter. Therefore, in this section, we first briefly introduce the MoE and the CLIP-Adapter frameworks, followed by a detailed description of the proposed MTL method, namely CLIP-MoA.

A. MoE

The MoE architecture consists of a set of independent expert networks and a gating network. Following the "divide and conquer" principle, the input features are partitioned into distinct feature subspaces, each of which is assigned to a specific expert selected by the gating network according to the task characteristics. The gating network takes task-specific features as input and generates a set of weights indicating the contribution of each expert to processing the input data, thereby integrating knowledge from multiple experts. The output dimension of the gating network equals the number of experts. Once the knowledge from all experts is fused, a unified output is produced, which is then fed into the top tower to generate the final result.

Given the input task feature x, the number of experts m, and the number of tasks M, the MoE model can be expressed as follows:

$$y = h(f(x)). \tag{1}$$

Here, y denotes the output of the top tower $h(\cdot)$; $f(x)$ represents the weighted sum of the outputs from n experts' outputs $f_i(x)(i = 1, 2, \ldots, n)$, as expressed in the following equation:

$$f(x) = \sum_{i=1}^{n} (g(x)_i f_i(x)). \tag{2}$$

Here, $g(x)$ denotes the gating network, where $\sum_{i=1}^{m} g(x)_i = 1$. The gating network typically comprises a single layer followed by the Softmax activation function, as defined in the following equation:

$$g(x) = \text{Softmax}\big(W_g x\big). \tag{3}$$

Here, $W_g \in \mathbb{R}^{m \times d}$ is a learnable weight matrix that encodes the contribution of each expert to the task, and d denotes the dimension of the input features.

By leveraging the gating network, MoE enables the development of specialized experts during the training process, while facilitating knowledge sharing among expert models during inference. Integrating MoE into large-scale models can enhance the overall performance in terms of generalization, representation, and learning capabilities without a significant increase in computational costs.

B. CLIP-Adapter

CLIP-Adapter is a simple yet efficient fine-tuning method that incorporates a residual-connected feature adapter into the visual or language branch of the frozen CLIP model, as illustrated in Fig. 3, enabling improved adaptation to downstream tasks such as image classification.

Specifically, for a downstream dataset with K-category image classification, a visual feature adapter $A^v(\cdot)$ and a text feature adapter $A^t(\cdot)$ are introduced into the visual and text branches of CLIP, respectively, to refine the original visual features f and text features $W = \{W_i | i = 1, 2, \ldots, K\}$. Here, W_i is derived from the hard prompt H, such as "a photo of a $\{C_i\}$" using the text encoder BERT, as expressed in the following equation:

$$W_i = \text{BERT}(\text{Tokenizer}[H; C_i]) \tag{4}$$

where, C_i represents the textual description of the ith category name.

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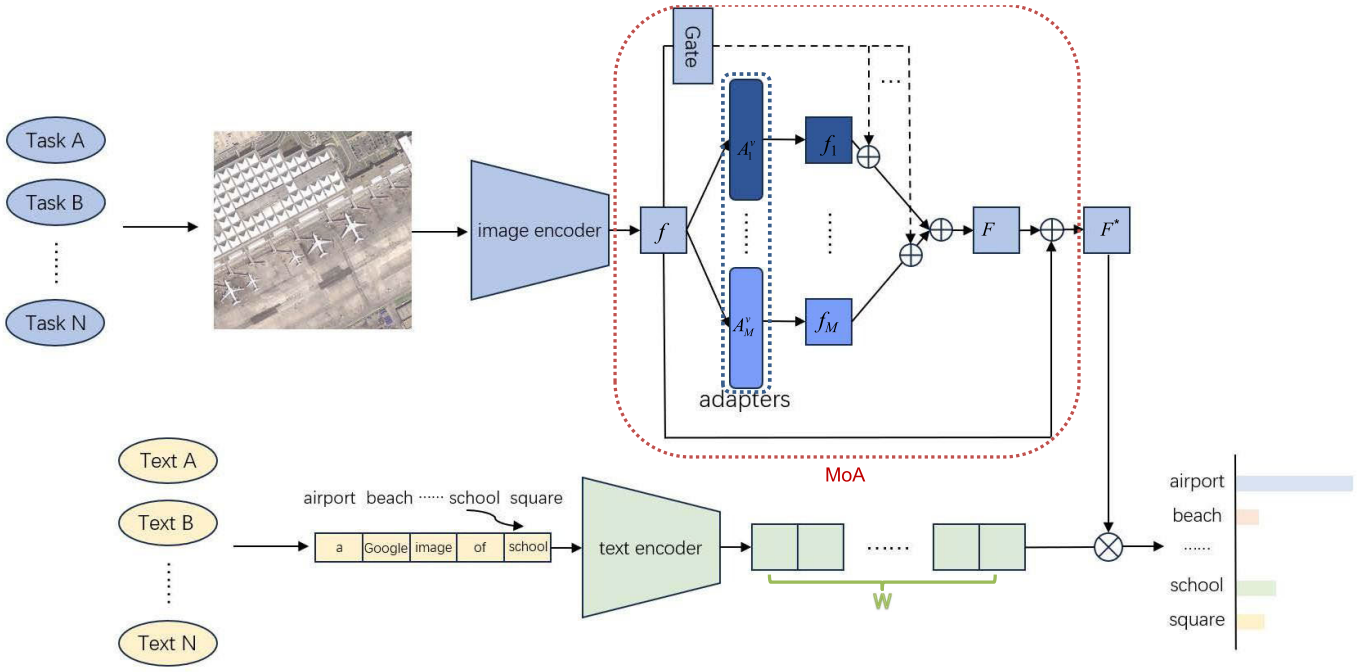


Fig. 4. Overall structure of CLIP-MoA. The MoA module, enclosed within the red dashed box, comprises multiple independent learnable adapters, a learnable gating network, and a residual connection, following a structure similar to MoE, where each adapter is identical to the Adapter in CLIP-Adapter and serves as an expert. The number of adapters is configurable based on the number of datasets, and in this study, it is set to equal the number of datasets.

Each adapter comprises two linear layer transformations with learnable parameters w_1^v and w_2^v for the image adapter and w_1^t and w_2^t for the text adapter, as defined in following equations:

$$A^v(f) = \mathrm{ReLU}\left(f^T w_1^v\right) w_2^v \tag{5}$$

$$A^t(W) = \mathrm{ReLU}\left(W^T w_1^t\right) w_2^t. \tag{6}$$

The new knowledge obtained through network fine-tuning is integrated with the original image feature f and text feature W through residual connections

$$f^* = \alpha A^v(f)^T + (1-\alpha) f \tag{7}$$

$$W^* = \beta A^t(W)^T + (1-\beta) W. \tag{8}$$

This residual fusion strategy enables the model to effectively adapt to the specific demands of new tasks while preserving pre-trained knowledge without incurring a substantial computational overhead. The resulting category probability vector is computed using the following equation:

$$p_i(f^*) = \frac{\exp\left(\frac{\left(W_i^*\right)^T f^*}{\tau}\right)}{\sum_{j=1}^{K} \exp\left(\frac{\left(W_j^*\right)^T f^*}{\tau}\right)}. \tag{9}$$

The image category is determined by identifying the class with the highest probability, expressed as $\hat{i} = \mathrm{argmax}_i\, p_i(f^*)$.

The adapter can be integrated into either the visual or text branch, or both branches simultaneously. Fig. 3 illustrates a specific implementation of CLIP-Adapter, in which a single learnable image feature adapter is incorporated into the visual branch of CLIP.

C. CLIP-MoA

Building upon the foundational concepts of CLIP-Adapter and MoEs, this article proposes CLIP-MoA, as illustrated in Fig. 4. CLIP-MoA incorporates a learnable, residual-connected MoA module into the visual branch of the CLIP framework. The MoA module establishes an analogy between adapters and expert modules in the MoE framework, seamlessly integrating the adapter concept with the MoE methodology. The image features from the frozen backbone are divided into distinct feature subspaces, each assigned to a specific adapter (expert) selected by the gating network based on the task characteristics. The gating network takes task-specific features as input and produces a set of weights that reflect the contribution of each adapter to processing the input data, thereby combining knowledge from multiple adapters. To mitigate catastrophic forgetting, the original general semantic image features are fused with the gating-weighted sum of the adapter outputs through a residual connection. Once the knowledge from all adapters and the original image features is fused, a unified output is generated, which is then used to calculate the probability of predicting the class associated with the input image. The details will be introduced in Sections III-C1 and III-C2.

1) MoA: Inspired by MoE, the MoA module adopts a similar architecture, which integrates multiple adapters through a gating mechanism and preserves the input features via a residual connection between the adapter inputs and the gate outputs. Here, the number of adapters is set to match the number of datasets. Each adapter is designed identically to the adapter in CLIP-Adapter, as formally defined in (5).

Specifically, let M denotes the number of image classification datasets(tasks), N represents the total number of samples from all datasets in a batch, and K signifies the total number

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of classes from all datasets. Given a set of multitask input images $\mathbb{I}$ and a set of class names $\mathbb{C}$ as follows:

$$\mathbb{I} = \left\{ I_{ij} | j = 1, 2, \ldots, n_i;\ i = 1, 2, \ldots, M;\ \sum_{i=1}^{M} n_i = N \right\}$$

$$\mathbb{C} = \left\{ C_{ij} | j = 1, 2, \ldots, k_i;\ i = 1, 2, \ldots, M;\ \sum_{i=1}^{M} k_i = K \right\}$$

where n_i denotes the number of samples I_{ij} from the ith dataset, and k_i represents the number of class names $C_{i,j}$ for the ith dataset. The image feature f and classifier weight $\mathbb{W} = \{W_{ij} | j = 1, 2, \ldots, k_i; i = 1, 2, \ldots, M\}$ are derived from the original CLIP backbone, where W_{ij} denotes the weight of the jth class in the ith task, as computed by the following equation:

$$W_{ij} = \text{BERT}\big(\text{Tokenizer}\big[H; C_{ij}\big]\big). \tag{10}$$

W_{ij} is subject to fine-tuning through the MoA module. In this work, W_{ij} remains frozen, and solely the image feature f is fed into M adapters $\{A_i^v(\cdot) | i = 1, 2, \ldots, M\}$ to generate M task-specific image features $\{f_i(\cdot) | i = 1, 2, \ldots, M\}$, as defined in the following equation:

$$f_i = A_i^v(f). \tag{11}$$

Here, $A_i^v(f)$ is consistent with (5) in CLIP-Adapter.

The M features $\{f_i(\cdot) | i = 1, 2, \ldots, M\}$, each of which is weighted by the gate network $g(f)$, which is fine-tuned online, are aggregated to derive the estimated feature F

$$F = \sum_{i=1}^{M} \big(g(f)_i \cdot (f_i)^T\big). \tag{12}$$

Here, $g(f)_i$ is consistent with (3) in the MoE framework.

To mitigate the risk of catastrophic forgetting of the original CLIP knowledge and to improve the robustness of CLIP-MoA, a residual connection is employed to dynamically integrate the fine-tuned knowledge with the original knowledge from the frozen CLIP backbone. A constant value α is introduced as the residual ratio to regulate and preserve the proportion of original knowledge, thereby enhancing the model's performance. Following the residual-connected MoA module, the resulting image feature F^* is computed as defined in the following equation:

$$F^* = \alpha f + (1 - \alpha) F. \tag{13}$$

2) Multitask Learning: Upon obtaining the new image features F^*, the following equation is utilized to calculate the probability p_{ij} that the input image belongs to the jth class of the ith task:

$$p_{ij}\big(F^*\big) = \frac{\exp\left(\frac{(W_{ij})^T F^*}{\tau}\right)}{\sum_{i=1}^{M} \sum_{j=1}^{k_i} \exp\left(\frac{(W_{ij})^T F^*}{\tau}\right)}. \tag{14}$$

Here, we set $\tau = 1$. The training objective is to predict the class associated with the input image by selecting the class $\hat{j}$ within the task $\hat{i}$ that maximizes probability p_{ij}, i.e., $\{\hat{i}, \hat{j}\} = \arg\max_{\{i,j\}} p_{ij}$.

For the learnable MoA module, the multitask loss function is defined as the average cross-entropy loss computed over N samples across M tasks, as expressed in the following equation:

$$L(\theta) = -\frac{1}{N} \sum_{i=1}^{M} \sum_{j=1}^{k_i} \sum_{s=1}^{n_{ij}} y_{ijs} \log \hat{y}_{ijs}. \tag{15}$$

Here, k_i denotes the number of classes in task i, n_{ij} represents the number of training samples for class j in task i, and N is defined as the total number of training samples across M tasks; y_{ijs} takes the value 1 if $i = \hat{i}$ and $j = \hat{j}$; otherwise $y_{ijs} = 0$; $\hat{y}_{ijs} = p_{ij}$ denotes the predicted probability that sample s belongs to class j of task i; $\theta = \{w_1^v, w_2^v, W_g\}$ encapsulates all learnable parameters associated with the adapters and the gate.

IV. Experiments

To evaluate the effectiveness of the CLIP-MoA model, a comprehensive set of comparison and ablation experiments was performed, respectively, on a multitask dataset comprising 11 RS image classification datasets. The proposed model was compared against four baseline models: zero-shot CLIP, linear probe CLIP, CLIP-Adapter, and CoOp.

A. Datasets

A total of 11 RS image datasets were selected for evaluation: AID [44], CLRS [45], NWPU_RESISC45 [46], PatternNet [47], RS_C11 [48], RS_Images_2800 [49], RSI_cb128, RSI_cb256 [50], SIRI_WHU [51], UC_Merced Land_Use [52], and WHU_rs19 [53]. Since the data sources of these datasets vary, they collectively form a diverse and comprehensive multitask dataset. The number of classes for each dataset and the corresponding prompt templates are summarized in Table I. A detailed description of four representative datasets is provided below.

1) UC_Merced Land_Use Dataset: The UC Merced Land-Use dataset is sourced from the United States Geological Survey's (USGS) National Map Urban Area Imagery collection. The dataset comprises 2100 aerial scene images, with each image belonging to one of 21 land-use scene categories, as illustrated in Fig. 5, encompassing diverse land-use scenes from urban and rural areas across the United States. These categories encompass common land-use types found in urban and rural areas, including farmland, forest, industrial areas, among others. Each category comprises 100 images, each with a resolution of 256 × 256 pixels. The publicly available images feature a pixel resolution of 1 ft, offering high levels of detail and clarity. However, the presence of similarities between certain categories introduces additional challenges for classification tasks, despite the dataset's diversity.

2) AID Dataset: AID dataset is compiled and annotated from imagery obtained through Google Earth. It is a new large-scale aerial image dataset. It includes a wide range of scene types and images. Specifically, it consists of 10 000 fixed-sized images of 600 × 600 pixels across 30 categories, as shown in Fig. 6. Within these categories, there are

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TABLE I
DATASET INFORMATION

Dataset Name	Number of Classes	Class Names	Number of Test Samples	Prompt Template
AID	30	Airport, BareLand, BaseballField, ...	3,000	a Google Earth Image of {ClsName}.
CLRS	25	Bridge, Commercial, Desert, ...	4,500	a Google Earth Image of {ClsName}.
NWPU	45	Chaparral, Church, Circular Farmland, ...	9,450	a Google Earth Image of {ClsName}.
PatternNet	30	Golf Course, Intersection, Mobile Home Park,...	9,120	a Google Earth Image of {ClsName}.
RS_C11	25	Dense forest, Grassland, Harbor, ...	370	a Google Earth Image of {ClsName}.
RS_Images_2800	18	Grass, Field, Industry, ...	840	a Google Earth Image of {ClsName}.
RSI_cb128	35	Construction Land, CultivatedLand, Other Land, ...	11,012	a Google Earth Image of a {ClsName}.
RSI_cb256	45	Construction Land, CultivatedLand, Other Land, ...	7,424	a Google Earth Image of {ClsName}.
SIRI_WHU	12	Agriculture, Commercial, Industrial, ...	720	a Google Earth Image of {ClsName}.
UC_Merced Land_Use	21	Freeway, River, Runway, ...	630	a Remote Sensing Image of {ClsName}.
WHU_rs19	19	Park, Parking, Pond, ...	301	a Remote Sensing Image of {ClsName}.

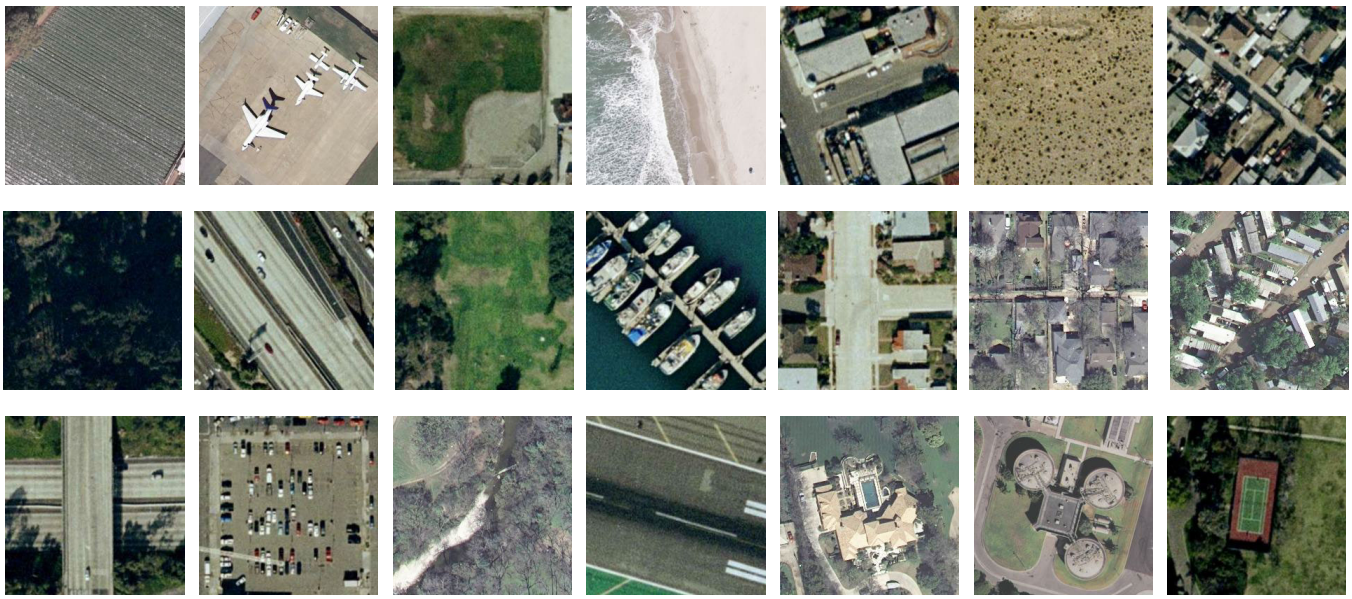


Fig. 5. Example images of UC Merced Land-Use dataset. (Top left) to (Bottom right) Agriculture, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis court.

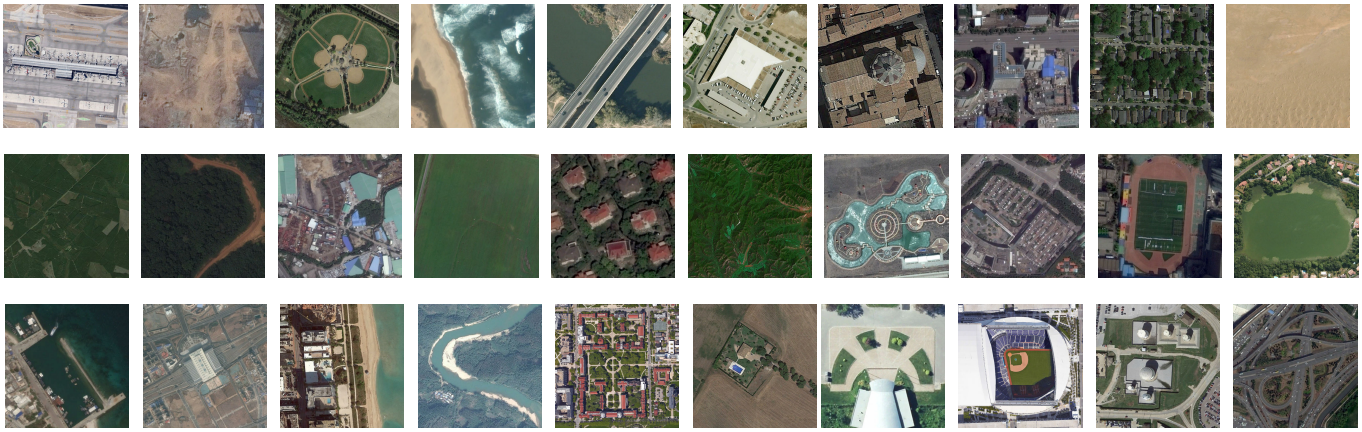


Fig. 6. Example images of AID dataset. (Top left) to (Bottom right) Airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct.

some similar classes that reduce inter-class variability, and the number of images varies from 220 to 420 for different scene types. Additionally, the AID dataset contains images captured at different times, seasons, and imaging conditions, from various countries and regions around the world, such as China, the United States, the U.K., France, Italy, and others.

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Therefore, compared to the aforementioned UC_Merced Land_Use dataset, the AID dataset is a more challenging dataset.

3) NWPU_RESISC45 Dataset: The NWPU_RESISC45 dataset is widely utilized for land object classification and recognition tasks in the RS domain. Developed by Northwestern Polytechnical University in China, the dataset comprises images captured from high-resolution satellite imagery. Compared to the UC Merced Land Use and AID datasets, this dataset exhibits greater complexity, featuring 45 distinct categories of land objects, such as deserts, forests, grasslands, farmland, lakes, roads, among others. Each category contains 700 images, yielding a total of 31 500 images. The images have a resolution of 256 × 256 pixels and are stored as RGB color images. The spatial resolution ranges from approximately 30 to 0.2 m/pixel. Additionally, the images encompass over 100 countries and regions globally.

4) PatternNet Dataset: The PatternNet dataset was introduced in 2018 by a research team from Wuhan University, aiming to provide an open benchmark dataset for RS image analysis and land object classification. The dataset comprises RS images across 38 distinct categories, including buildings, water bodies, forests, roads, and farmland. Each category includes approximately 2000 image samples, yielding a total of over 76 000 images. The images have a resolution of 256 × 256 pixels and are stored as RGB color images.

B. Training Details

To ensure the comparability of the experimental results with the baseline models, the batch size was set to 32 and the learning rate was fixed at 1×10^{-5}. ResNet-50 [5] was employed as the visual backbone (visual encoder) and and BERT was utilized as the text encoder (classifier weight generator). For text input, hard prompts aligned with CLIP-Adapter were adopted, as opposed to the soft prompts employed in CoOp. Experiments were conducted with various hard prompt templates, as shown in Table I, and the template yielding the best performance was selected for CLIP-Adapter and CLIP-MoA.

For CLIP-Adapter, to achieve optimal classification performance, a parameter search was conducted for each dataset to determine the optimal configuration of adjustable parameters, namely, the residual ratio α for the feature adapter in the residual connection of the backbone and the hidden embedding dimension ρ of the visual bottleneck layer. For instance, for the AID dataset, the optimal parameter configuration was $\alpha = 0.9$ and $\rho = 256$.

Since CoOp employs continuous vectors, the class token can be placed at multiple candidate locations within the prompt template, including the beginning, middle, or end. In this work, the top-performing variant of CoOp was adopted, which positions the class token at the end of the prompt template with a length of 16 and shares the context across different classes.

Linear probe CLIP trains a linear classifier using high-quality pre-trained model features, achieving performance on par with state-of-the-art few-shot learning methods. This method employs a few-shot training approach, dynamically fine-tuning image features and classifier weights with residual connections, in contrast to our bottleneck adapter.

The few-shot evaluation protocol employed in CoOp and CLIP-Adapter was adopted, with experiments conducted using 1, 2, 4, 8, and 16 shots. The deployed models were evaluated, and the final results were determined based on the average performance across three runs.

A separate validation set was utilized for hyperparameter tuning. The validation set was randomly sampled from each dataset and excluded from the model training process. A fixed proportion of data (typically 20%) was randomly sampled from each dataset to form the validation set.

C. Performance Comparison and Analysis

Classification experiments were conducted using zero-shot CLIP, linear probe CLIP, CoOp, CLIP-Adapter, and CLIP-MoA across different shot settings on 11 RS image classification datasets. The experimental results are presented in Fig. 7. Based on the average accuracy across the 11 datasets, CLIP-MoA consistently outperforms the other four baseline models in terms of classification accuracy across all shot settings, highlighting its superior few-shot learning capabilities relative to CLIP-Adapter. Compared to the baseline models, CLIP-MoA demonstrates more pronounced advantages in 1-, 2-, and 4-shot settings compared to 8- or 16-shot settings. This suggests that CLIP-MoA achieves better performance than CLIP-Adapter under data-scarce conditions, exhibiting enhanced generalization capabilities.

Compared to zero-shot CLIP, the average accuracy across the 11 datasets indicates that CLIP-MoA achieves an improvement of nearly 20%. For individual datasets, while CLIP-MoA consistently outperforms zero-shot CLIP across all datasets, the other three baseline models may underperform zero-shot CLIP in 1-shot settings on specific datasets, such as the CLRS and NWPU_RESISC45 datasets. These findings further confirm that CLIP-MoA exhibits stronger generalization capabilities compared to CLIP-Adapter in low-shot settings.

Compared to linear probe CLIP, the average accuracy across the 11 datasets reveals that CLIP-MoA maintains a significant accuracy advantage, especially in low-shot settings like 1- and 2-shot, with a margin of approximately 10%–15%. On certain datasets, the performance gap between CLIP-MoA and linear probe CLIP is even more pronounced; for instance, CLIP-MoA outperforms linear probe CLIP by over 20% in 1-shot on the CLRS dataset. However, linear probe CLIP demonstrates exceptional performance in the specific task of RS scene classification, owing to the robust pre-trained CLIP space, which is expected.

Compared to zero-shot CLIP, CoOp already exhibits a significant improvement. However, relative to CLIP-Adapter and CLIP-MoA, a performance gap remains. This further validates that the use of hard prompts in CLIP-Adapter and CLIP-MoA, combined with fine-tuning of pretrained VLMs and integration of prior knowledge via residual connections, yields superior performance compared to CoOp with prompt tuning.

Compared to CLIP-Adapter, which already achieves near-optimal results on average across the 11 datasets,

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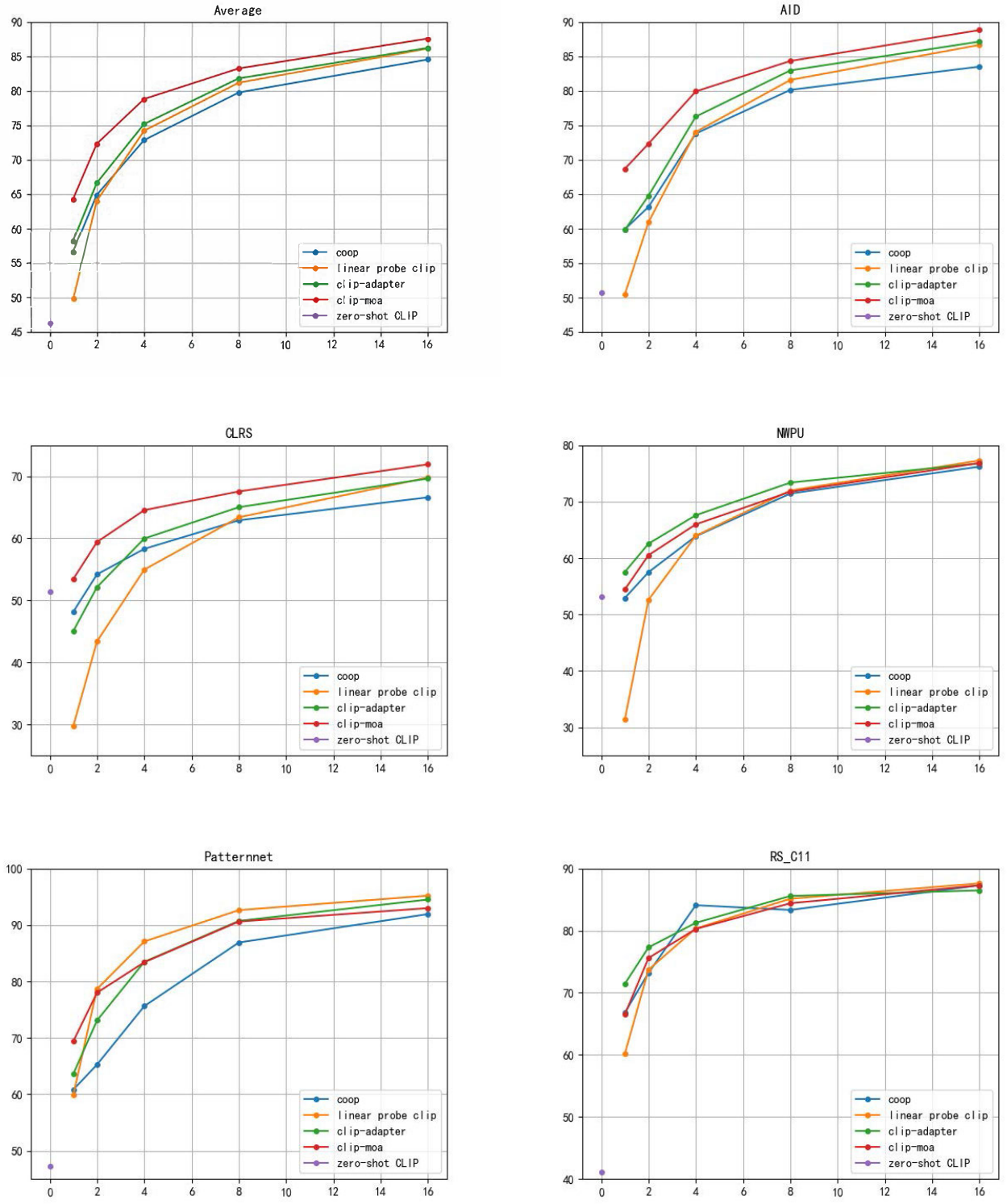


Fig. 7. Classification accuracy under different shots on 11 datasets: the horizontal axis represents the number of K for shots, and the vertical axis represents the classification accuracy. The results show that CLIP-MoA has achieved significantly better results with a universal set of parameters for different datasets.

CLIP-MoA achieves higher accuracy, and further demonstrates its superiority, fully corroborating the earlier claims. It is also observed that on the NWPU_RESISC45 dataset, CLIP-Adapter outperforms CLIP-MoA. This can be attributed to the inherent complexity of the NWPU_RESISC45 dataset and the larger number of trainable parameters in CLIP-MoA compared to CLIP-Adapter. In scenarios with limited training samples, the higher number of parameters may result in slight overfitting, thereby impacting the performance of CLIP-MoA on the NWPU_RESISC45 dataset.

Furthermore, CLIP-MoA demonstrates significant advantages over CLIP-Adapter. The results of CLIP-Adapter are

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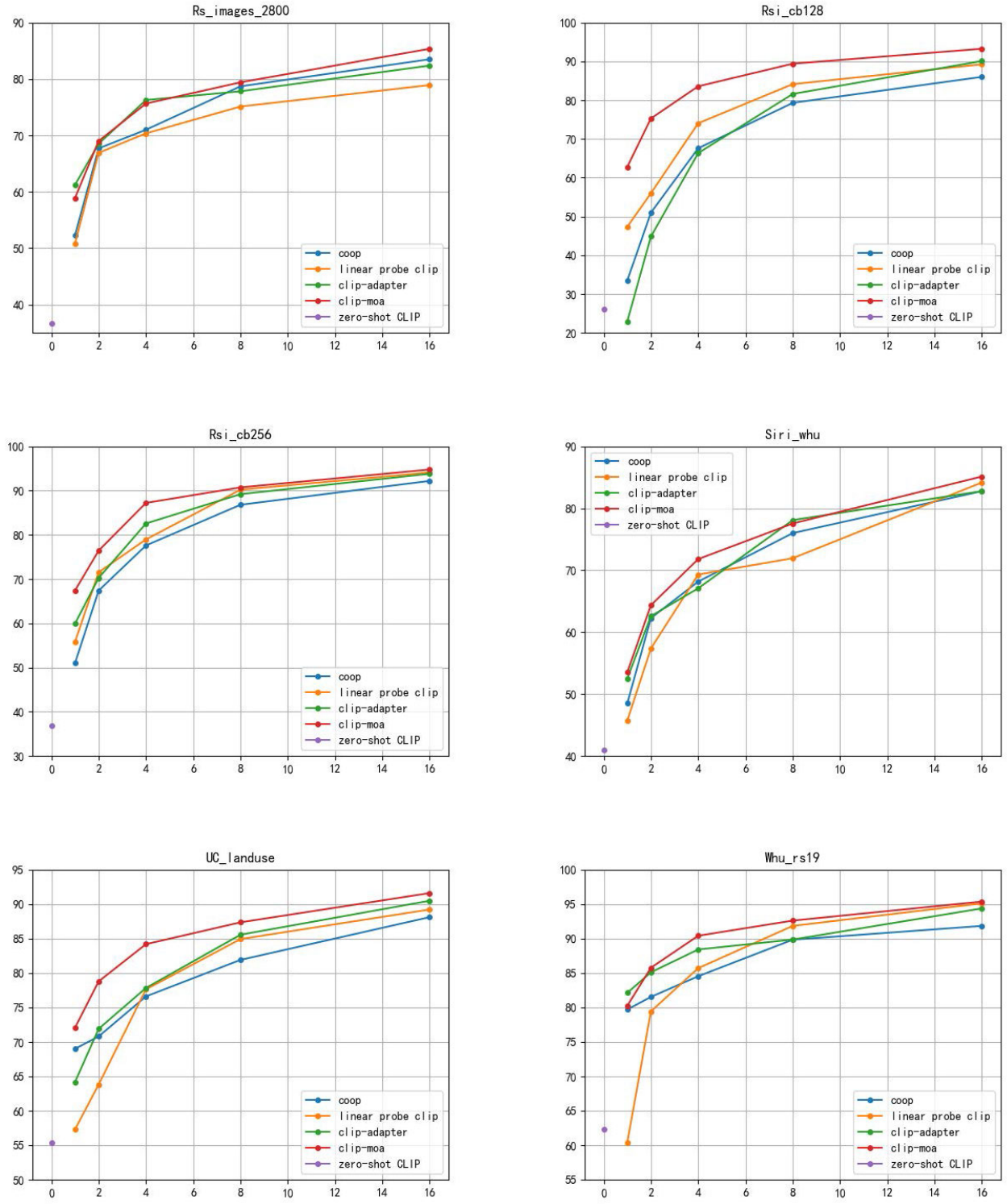


Fig. 7. *(Continued.)* Classification accuracy under different shots on 11 datasets: the horizontal axis represents the number of K for shots, and the vertical axis represents the classification accuracy. The results show that CLIP-MoA has achieved significantly better results with a universal set of parameters for different datasets.

derived from individually tuning and selecting the optimal parameter configuration for each dataset. In contrast, CLIP-MoA employs a unified parameter configuration across all datasets, achieving superior performance compared to CLIP-Adapter.

This underscores the strong transfer learning ability of CLIP-MoA, which adapts to multiple downstream datasets using a single model, thereby conserving significant resources while achieving better performance. This capability stems primarily from the MTL architecture of CLIP-MoA and the

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inherent characteristics of the pre-trained CLIP model. On one hand, CLIP-MoA preserves the original features of CLIP via residual connections, allowing it to rapidly adapt to new tasks without requiring training from scratch. On the other hand, the MTL mechanism dynamically integrates the outputs of multiple adapters using an MoA module and a gating network, where each adapter corresponds to a specific task or dataset. This enables the model to leverage shared knowledge across different tasks or datasets while maintaining task-specific characteristics.

In few-shot scenarios, such as 1- and 2-shot, CLIP-MoA achieves high accuracy with minimal fine-tuning, demonstrating its capability to rapidly adapt to new tasks with limited data. Experimental results demonstrate that CLIP-MoA outperforms individually trained models on 11 RS datasets following unified hyperparameter tuning, validating its effectiveness in cross-task transfer for RS image classification.

D. Results Visualization

The t-SNE visualization approach [54] as CLIP-Adapter was employed to visualize the results of CLIP-Adapter and CLIP-MoA across various RS datasets. The visualization results for the selected eight datasets are presented in Fig. 8, with different colors indicating distinct categories. For instance, in the AID dataset, the colors correspond to categories including airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, and desert. For datasets with fewer than ten categories, such as RS_Images_2800, the number of colors is adjusted to align with the number of categories.

Fig. 8 illustrates the clear separation of image features across different classes achieved by CLIP-MoA. Although CLIP-Adapter achieves exceptional performance on certain datasets, such as AID, NWPU_RESISC45, and PatternNet, demonstrating clear separation of image features across different classes, some individual classes exhibit less distinct separation. For instance, in the AID dataset, the categories of airports, beaches, and churches are less distinguishable, as their features are intermixed. However, the visualization results of CLIP-MoA reveal that the separation of these three categories is more distinct compared to CLIP-Adapter. Furthermore, the visualization results collectively demonstrate that CLIP-MoA outperforms CLIP-Adapter across multiple datasets, beyond just a few specific classes, as exemplified by the RS_Images_2800 dataset. In conclusion, the visualization results validate that CLIP-MoA can more effectively learn features in few-shot settings and achieve superior results for certain challenging classes.

E. Ablation Studies and Hyperparameter Sensitivity Analysis

This section will use the single parameter sensitivity analysis method to analyze the impact of hyperparameters such as bottleneck layer dimension (ρ), residual ratio (α), and number of adapters on the classification accuracy of CLIP-MoA. The ablation studies were implemented under 16-shot training conditions across multitask datasets. By varying one parameter while fixing the others, we observed changes in model performance and analyze the functions of each hyperparameter on the model performance. It can provide important references for designing model hyperparameters.

1) Dimension of Bottleneck Layer: The bottleneck layer dimension (ρ) regulates the feature compression ratio within the adapter module's architecture. With fixed the other hyperparameters, the bottleneck layer dimension ρ was evaluated across a range of 128–512 in increments of 128, enabling comprehensive analysis of its influence on computational efficiency and model accuracy. The experimental results are shown in Table II.

Extensive experimental validation revealed that a bottleneck dimension of $\rho = 256$ yields optimal performance across diverse datasets when $\alpha = 0.6$. Suboptimal ρ values demonstrated a clear trade-off: reduced dimensions risk feature information degradation, whereas excessive dimensions escalate computational overhead without commensurate performance gains. This finding indicates that specialized DA necessitates greater reliance on newly acquired knowledge than on preexisting CLIP representations.

2) Residual Ratio: The residual ratio (α) modulates the feature integration between the foundational CLIP representations and the adapter-derived feature mappings. The residual ratio α was modulated from 0 to 1 in 0.1 increments, with other hyperparameters fixed at the optimal values, to quantify its role in balancing original and adapted feature contributions. Empirical results presented in Table II.

Empirical analysis identified an optimal residual ratio of $\alpha = 0.6$ when $\rho = 256$, consistently delivering superior performance across heterogeneous datasets. This finding implies that domain-specific feature adaptation through the adapter module carries greater significance than the original CLIP feature representations for target tasks. Deviation from the optimal α value manifests in performance degradation: excessive ratios bias the model toward adapter features at the expense of foundational CLIP knowledge, while insufficient ratios underutilize the adapter's learning potential. The boundary cases reveal distinct operational modes: at $\alpha = 0$, the system reduces to zero-shot CLIP by exclusively utilizing pre-trained knowledge, while $\alpha = 1$ represents complete dependence on domain-specific adaptations. Consistent with findings in CLIP-Adapter, both extreme configurations exhibit heightened susceptibility to overfitting, resulting in measurable degradation of classification accuracy.

3) Number of Adapters: The number of adapters determines the model's ability to learn task-specific feature representations. A systematic ablation study was conducted to quantify the relationship between adapter quantity and classification accuracy. The adapter number was systematically varied from 7 to 15 in steps of one, maintaining constant values for all other hyperparameters. The experimental results are visualized in Fig. 9, illustrating the performance trends across different adapter configurations.

Quantitative analysis reveals that optimal performance is achieved when the number of adapters exactly matches the number of target tasks (datasets). An insufficient or excessive number of adapters may lead to overfitting or underfitting, thereby reducing classification accuracy. This optimal

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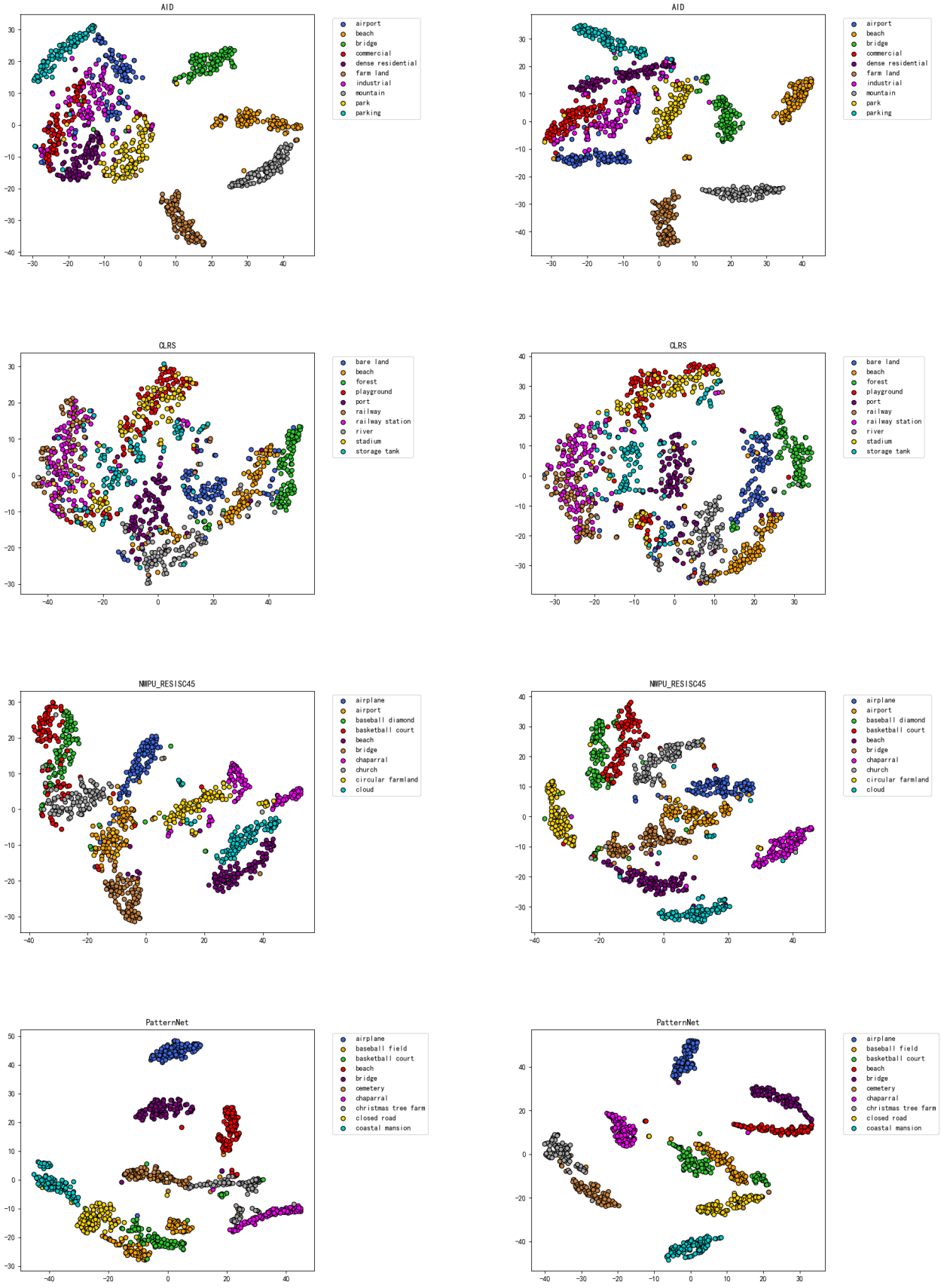


Fig. 8. (Left) Classification visualization results on 11 datasets of CLIP-Adapter and (right) CLIP-MoA through t-SNE.

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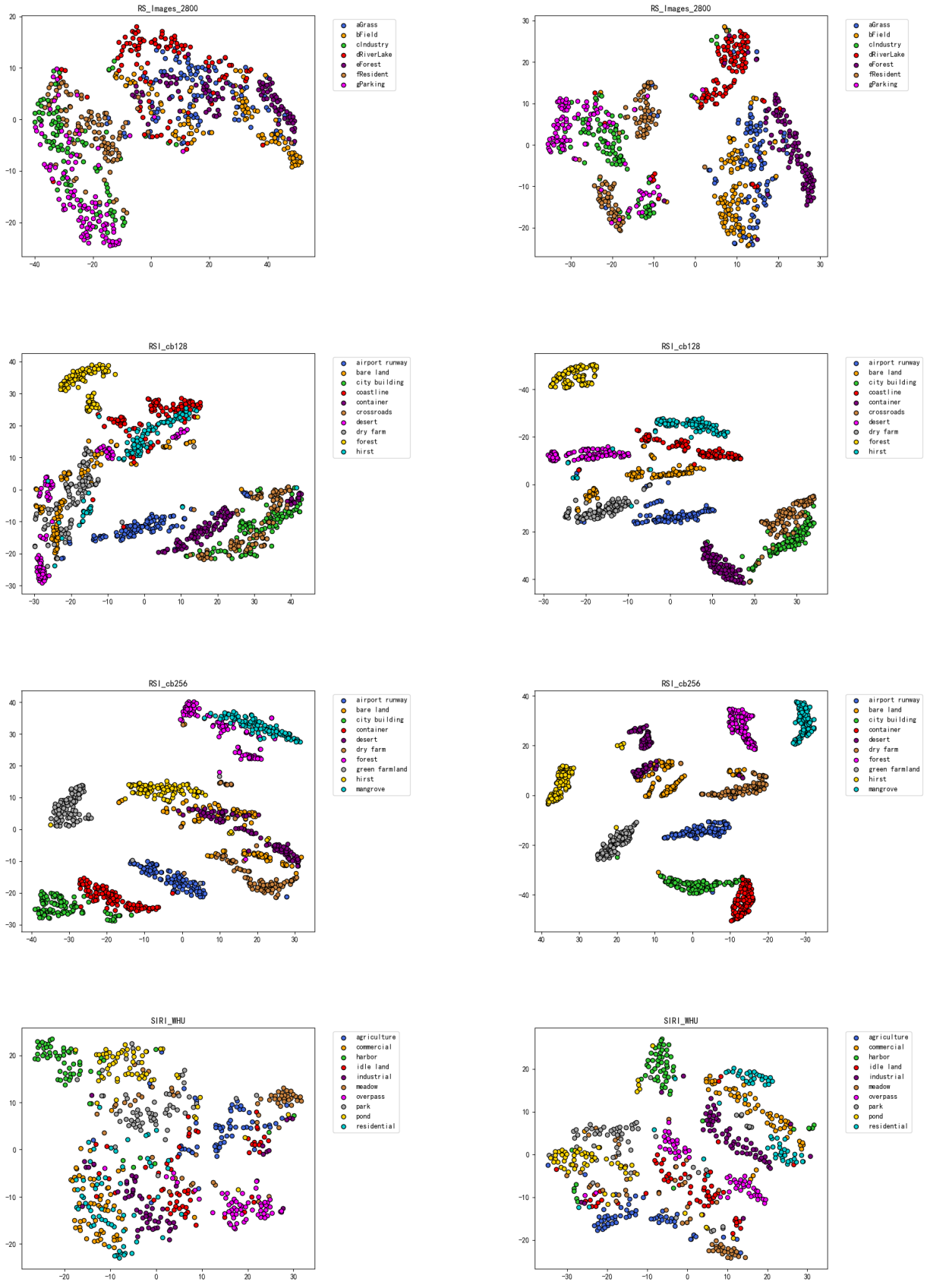


Fig. 8. *(Continued.)* (Left) Classification visualization results on 11 datasets of CLIP-Adapter and (right) CLIP-MoA through t-SNE.

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TABLE II
AVERAGE ACCURACY ACROSS 11 DATASETS UNDER DIFFERENT BOTTLENECK LAYER DIMENSIONS ρ AND RESIDUAL RATIO α FOR 16-SHOT TRAINING SETTINGS

Dimension(ρ)	Ratio(α)										
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
128	86.84%	81.95%	86.53%	86.38%	86.52%	86.82%	86.75%	86.59%	86.85%	86.77%	86.39%
256	86.82%	82.03%	85.95%	86.58%	87.02%	87.24%	**87.57%**	87.14%	87.52%	87.00%	86.84%
512	86.42%	81.88%	86.82%	86.51%	86.76%	86.77%	86.81%	86.84%	86.92%	86.63%	86.43%

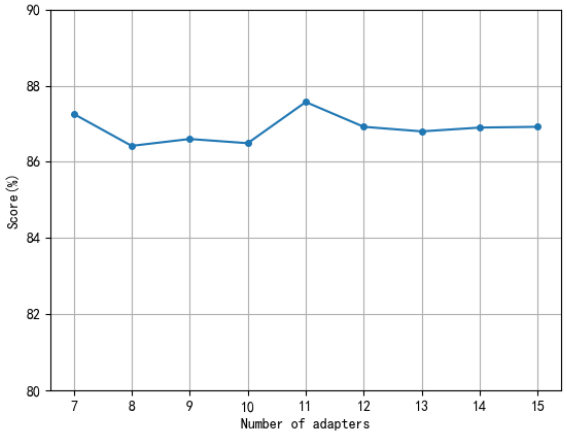


Fig. 9. Average classification accuracy across 11 datasets under different number of adapters.

TABLE III
AVERAGE CLASSIFICATION ACCURACY OF CLIP-MOA AND CLIP-MMOA ACROSS 11 RS DATASETS

CLIP-MoA	**CLIP-MMoA**
87.57%	86.83%

configuration can be explained by two key factors: 1) task-specific adapters preserve task-specific features and enable comprehensive feature learning and representation and 2) the parameter space achieves an optimal balance, promoting precise and effective inter-task information sharing while preserving model generalization capabilities.

4) Number of Gates: To investigate the impact of gating mechanisms on classification accuracy, we developed a multigate MoAs (MMoA) architecture, inspired by MMoE, where each adapter is associated with a dedicated gating network. This architecture was implemented within CLIP's visual pathway, termed CLIP-MMoA. A comprehensive comparative analysis was conducted between CLIP-MoA and CLIP-MMoA architectures. The experimental results, presented in Table III, demonstrate that CLIP-MMoA exhibits inferior classification accuracy compared to CLIP-MoA under conditions of high model complexity and limited sample availability.

This performance degradation can be attributed to MMoA's multigate architecture, which substantially increases model parameterization, potentially inducing overfitting and excessive noise sensitivity in data-scarce scenarios, thereby compromising the model's ability to capture generalizable patterns. Furthermore, the independent gating mechanism restricts cross-task information flow, adversely impacting the model's capacity to generalize to novel tasks. In contrast, MoA's shared gating architecture achieves parameter efficiency, effectively mitigating overfitting risks while facilitating inter-task knowledge transfer through its unified gating network, thereby enabling more robust task relationship modeling. This inherent information-sharing mechanism empowers the model to leverage established knowledge representations when encountering novel tasks, significantly enhancing its generalization capabilities.

In summary, the ablation experimental results yielded several significant findings: optimal model performance across diverse datasets was achieved when: 1) the number of adapters matched the number of tasks; 2) the bottleneck layer dimension was set to $\rho = 256$; and 3) the residual ratio was $\alpha = 0.6$, highlighting the importance of a balanced hyperparameter configuration.

V. CONCLUSION

We present CLIP-MoA, a novel MTL framework for pre-trained vision-language models, specifically CLIP. Inspired by MoE, we developed the MoAs module, which employs multiple learnable task-specific adapters with a gating mechanism for adaptive feature integration and residual connections for balanced knowledge fusion between original and newly learned representations. The MoA module was strategically integrated into CLIP's visual pathway to create the CLIP-MoA architecture. CLIP-MoA possesses several benefits more appealing than prior arts. First, CLIP-MoA introduces MoA module to increase the CLIP's universality and efficiency without updating its original pre-trained parameters in RS image classification domain. Second, MoA can be plug-and-play in different Transformers and scalable to many visual tasks. Third, CLIP-MoA was evaluated on a comprehensive multitask RS image classification benchmark, comprising 11 distinct datasets specifically curated for this study. Extensive experimental results demonstrate CLIP-MoA's superior performance over zero-shot CLIP, linear probe CLIP, CLIP-Adapter, and CoOP across various few-shot learning scenarios. Comprehensive ablation studies further validate the optimality of our hyperparameter configurations and the efficiency of the architectural design. However, selecting a set of optimal hyperparameters is somewhat time consuming. Therefore, a future work will focus on automated machine learning techniques for hyperparameter tuning.

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